

Advanced Applied Econometrics

Static discrete choice with market-level data

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Organisation

- ▶ My part has three problem sets
- ▶ The third problem set is graded.
- ▶ June 19: recap demand estimation, discuss details, practical session (BLP)
- ▶ Read (again) Berry et al. (1995) and Conlon and Gortemaker (2020)

Plan for today

- ▶ Structural econometrics in Industrial Organization
- ▶ Demand for differentiated products: Berry (1994), Berry et al. (1995)
- ▶ Practical session: logit demand with aggregate data

Structural econometrics in Industrial Organization

- ▶ Highly recommended slides by Phil Haile on “Models, Measurement, and the Language of Empirical Economics” to read at:
<https://sites.google.com/view/philhaile/home/teaching>
- ▶ “Structural: estimate features of a data generating process (i.e., a model) that are (assumed to be) invariant to the policy changes or other counterfactuals of interest”

Structural econometrics in Industrial Organization

- ▶ IO applications: study the behavior of firms in markets, their strategic interactions, and the implications on profits and consumer welfare.

Recent policy discussions on inflation and markups

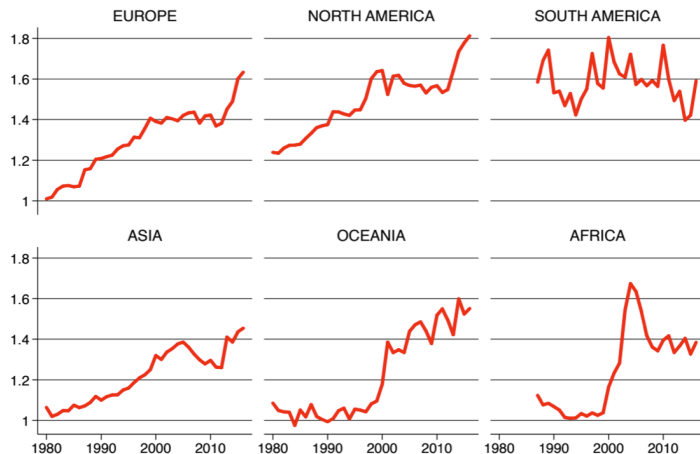
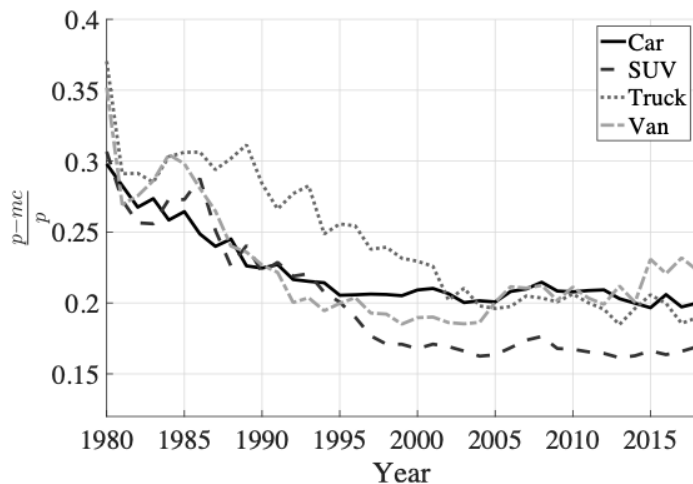


Figure 5: Global Regions

De Loecker, Jan and Jan Eeckhout (2021), "Global Market Power", working paper

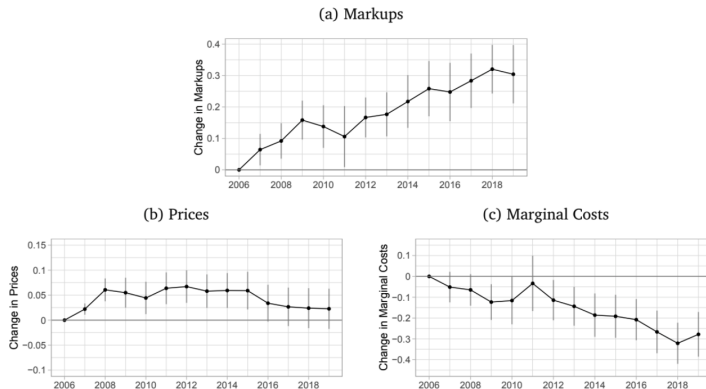
Recent policy discussions on inflation and markups



(a) Markups by Vehicle Style

Grieco, Paul, Charles Murry, and Ali Yurukoglu (2024), "The Evolution of Market Power in the US Auto Industry", Quarterly Journal of Economics

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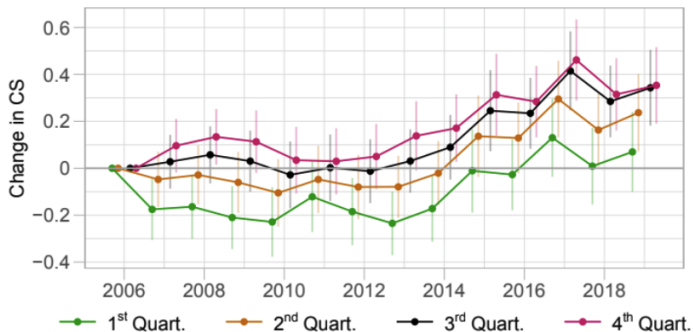


Notes: This figure shows coefficients and 95 percent confidence intervals of a regressions of the log of the Lerner index, real prices, and real marginal costs at the product-chain-DMA-quarter-year level on year dummies controlling for product-chain-DMA and quarter fixed effects. The year 2006 is the base category.

Döppler, Hendrik, Alexander MacKay, Nathan H. Miller, and Joel Stiebale (2024), “Rising Markups and the Role of Consumer Preferences”, *Journal of Political Economy*

Recent policy discussions on inflation and markups

Figure 10: Consumer Surplus Over Time By Income Group



Döpper, Hendrik, Alexander MacKay, Nathan H. Miller, and Joel Stiebale (2024), “Rising Markups and the Role of Consumer Preferences”, Journal of Political Economy

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Table 1
Analysis of the Industry Studies.

		Market Power?	Higher Prices?	Lower Costs? Better Quality?	Greater Scale?	Technological Change?
1	CPG	more	no	yes	n/a*	yes*
2	Cement	more	no	yes	yes	yes
3	Wholesalers	more	no	yes	yes	yes
4	Steel	less	no	yes	opposite	yes
5	Automobiles	less	yes*	yes	no	yes
6	Airlines	more	no	yes*	no	maybe*

$$\mu_{it} \equiv \frac{P_{it}}{MC_{it}} = \theta_{it}^V \frac{P_{it} Q_{it}}{P_{it}^V Q_{it}^V}$$

Cost minimization

vs.

$$\frac{P_{it} - MC_{it}}{P_{it}} = \frac{1}{\eta_{it}(P_{it}, \xi_{it})}$$

Profit-max pricing →
demand elasticities

Miller, Nathan H. (2025), "Industrial organization and The Rise of Market Power", International Journal of Industrial Organization 98, 103131.

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- ▶ Counterfactuals for ex-ante (welfare) evaluation of mergers, antitrust cases, regulation (e.g. price or entry barriers), subsidies, etc.

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 - ▶ Counterfactuals for ex-ante (welfare) evaluation of mergers, antitrust cases, regulation (e.g. price or entry barriers), subsidies, etc.
- Require a flexible, well-specified demand system

Typical IO Models

How is economic theory used as structure in IO applications?

1. Model of consumer behavior (demand), product differentiation
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Conduct

3. Equilibrium model of static competition, price (Bertrand), quantity (Cournot), bargaining, collusion, etc.
4. Equilibrium model of market entry-exit
5. Equilibrium model of dynamic competition, investment, advertising, quality, product characteristics, stores, etc.

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Typical Structure of IO Models

For this session:

1. Model of consumer behavior (demand)
2. Model of firms costs
3. Equilibrium model of static competition

Demand for differentiated products: discrete choice models

- ▶ Empirical observation:
 - ▶ Products mostly not homogenous
 - ▶ Firms typically set prices strategically
- ▶ Then, how to model and estimate demand for differentiated products
- ▶ Aggregation to market-level important. In 1990s, move from
 - ▶ representative consumer to
 - ▶ micro-modelling of consumer preferences.

Demand for differentiated products: discrete choice models

- ▶ One way: system of linear demand equations

$$q = D(p, x, \theta), \text{ with } D(p) = Ap$$

- ▶ J products
- ▶ Flexible demand system but number of parameters to be estimated: $J \times J$ matrix of price coefficients

Demand for differentiated products: discrete choice models

- ▶ Discrete choice models as models of product demand:
 - ▶ Characteristics space to reduce dimensionality (McFadden, 1974)
- ▶ Berry (1994): foundation for discrete choice demand estimation using market-level data, single-product firm oligopoly pricing
- ▶ Berry et al. (1995): first application of the proposed framework, multi-product firm oligopoly pricing, proof of contraction mapping

Discrete Choice Model of Demand

- ▶ Assume individual i receives utility U_{ij} from buying product $j \in J$
- ▶ The individual buys one unit of j : choice options are exhaustive and mutually exclusive
- ▶ A utility-maximizer chooses option j with probability: $s_{ij} = Pr(U_{ij} > U_{ik}) \forall k \neq j$
- ▶ Separating utility into observed V_{ij} and unobserved ϵ_{ij} components

$$\begin{aligned} s_{ij} &= Pr(V_{ij} + \epsilon_{ij} > V_{ik} + \epsilon_{ik}) \forall k \neq j \\ &= Pr(\epsilon_{ij} - \epsilon_{ik} > V_{ik} - V_{ij}) \forall k \neq j \end{aligned}$$

Discrete Choice Models as Models of Demand

- ▶ Defining $f(\epsilon_i)$ as J vector of i 's unobserved utility component,

$$s_{ij} = \int I(\epsilon_{ij} - \epsilon_{ik} > V_{ik} - V_{ij}) f(\epsilon_i) d\epsilon_i$$

- ▶ If ϵ_{ij} has extreme value type 1 distribution, consumer i chooses product j with probability:

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- ▶ EV1 $f(\epsilon_i = e^{-\epsilon_{ij}} e^{-e^{-\epsilon_{ij}}})$ is continuous with full support: smooth and bounded $s_{ij} \in (0, 1)$
→ continuously differentiable in V_{ij} and can rationalize any observed pattern in data
- ▶ Necessary normalizations for identification of V_{ij} : $V_{ik} = 0$ for one k and $\text{Var}(\epsilon_{ij} = \pi^2/6)$

Demand for differentiated products: Berry (1994)

- To aggregate, $\delta_j \equiv V_j$ is the mean utility of product j , common across all individuals. Consumer i receives linear utility:

$$u_{ij} = \delta_j + \varepsilon_{ij}$$

where $\delta_j = x_j\beta - \alpha p_j + \zeta_j$.

- Given the assumptions discussed earlier, the probability that consumer i chooses product j is:

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- ▶ Outside good is a natural candidate for $\delta_j = 0$. What is a reasonable market definition?

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- ▶ Market share inversion allows for linear estimation using
 - ▶ Fixed effects
 - ▶ Instruments - do we have good ones (for price)?

Instruments that have been used

- ▶ Good instruments might be the regressors
(optimal IV: derivative of the moment function with respect to the parameters)
- ▶ Cost shifters uncorrelated with the demand shock (rarely observed)...
- ▶ Bresnahan (1987), BLP (1995): Assume product characteristics exogenous. Use
 - ▶ observed product characteristics; sums of the same characteristics of other products offered by that firm; sums of the same characteristics of products offered by other firms
 - ▶ Intuition: closeness in characteristics space, in oligopoly markups decreases with the presence of good substitutes
- ▶ Gandhi and Houde (2019): Differentiation IVs
 - ▶ Polynomial of differences in observed product characteristics
- ▶ What about endogeneity of other product characteristics?

Demand for differentiated products: Berry (1994)

- ▶ Homogenous consumers (α, β) strong assumption
→ representative consumer up to i.i.d. ε_{ij}
- ▶ Leads to substitution patterns depending only on market shares

Unobserved preference heterogeneity

- ▶ Random coefficients: allow consumer to have heterogenous preferences over certain characteristics, e.g. β_i

$$u_{ij} = x_j \beta_i - \alpha p_j + \zeta_j + \varepsilon_{ij}$$

- ▶ Now can estimate completely flexible substitution patterns (market share derivatives depend on all characteristics in u_{ij})

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- ▶ Now can estimate completely flexible substitution patterns (market share derivatives depend on all characteristics in u_{ij})
- ▶ Berry (1994) inversion to obtain linear equation no longer possible
- ▶ Market shares, as we did not integrate out all individual-specific utility components:

$$s_j = \int \dots \int (s_{ij} dF_1(\alpha|\theta) \dots dF_{K+1}(\beta_k|\theta))$$

Unobserved preference heterogeneity

- ▶ Assume that individual i can be described by a vector of unobserved characteristics $(\nu_i, \varepsilon_{i0}, \dots, \varepsilon_{iJ})$.
- ▶ We can re-write consumer i 's utility:

$$u_{ij} = \delta_j(x_j, p_j, \xi_j; \theta_1) + \mu_{ij}(x_j, p_j, \nu_i, \theta_2) + \varepsilon_{ij}$$

→ Partition utility into mean utility δ_j and consumer-specific μ_{ij}

- ▶ Vector $\theta_1 = (\alpha, \beta)$ contains linear and $\theta_2 = (\sigma)$ nonlinear parameters

Unobserved preference heterogeneity

- ▶ For simplicity, assume that consumers only have heterogeneous preferences over one characteristic in x_j , so that $\theta_2 = (\sigma)$ is a scalar.
- ▶ Nonlinear part of consumer utility: $\mu_{ij}(x_j, p_j, v_i, \theta_2) = x_j v_i \sigma$
- ▶ The distribution of v_i needs to be assumed. For example, $N(0, 1)$.
- ▶ If consumer demographics are observed (e.g. age, income), empirical or fitted distributions

Simulation

- ▶ Idiosyncratic terms ε_{ij} integrated out (EV distribution assumption)
- ▶ But due to v_i no closed form solution for the integral
- ▶ Numerically invert the system of equation $s_j = s_j(x, p, \delta_j, \theta_2)$ to obtain δ_j given θ_2

Simulation

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- ▶ But due to v_i no closed form solution for the integral
- ▶ Numerically invert the system of equation $s_j = s_j(x, p, \delta_j, \theta_2)$ to obtain δ_j given θ_2
- ▶ Two numerical ingredients needed:
 - Numerical integration / simulation to compute market shares
 - Contraction mapping to compute δ_j

Integration to compute market shares

- ▶ Take s draws from the distribution of ν for each observation
- ▶ Given s draws of ν , σ , and δ , compute the market shares:

$$s_j = \frac{1}{ns} \sum_{i=1}^{ns} s_{ij} = \frac{1}{ns} \sum_1^{ns} \frac{\exp(\delta_j + x_j \nu_i \sigma)}{1 + \sum_{m=1}^J \exp(\delta_m + x_m \nu_i \sigma)}$$

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- ▶ Monte Carlo simulation
 - ▶ Pseudo-random draws
 - ▶ Quasi-random - e.g. Halton, modified Latin hypercube sampling
- ▶ Quadrature: Gauss-Hermite for (univariate) normal distribution, unbounded support

Contraction mapping for δ_j

- ▶ Given σ , compute vector δ that equates model predicted to observed market shares
- ▶ Contraction mapping

$$\delta^{h+1} = \delta^h + \ln(s) - \ln(s(p, x, \delta^h, \sigma)), h = 0, \dots, H,$$

where H is the smallest integer such that $\|\delta^H - \delta^{H-1}\|$ is smaller than some tolerance level, and δ^H is the approximation to δ .

- ▶ Compute vector $s(x, p, \delta, \sigma)$ at each iteration

Estimation

- ▶ Nested simulated GMM procedure
- ▶ IV estimation. Population moment conditions $E[Z\xi(\theta^*)] = 0$
- ▶ Using δ_j , computed at each parameter iteration in the GMM objective function, obtain:
$$\xi_j = \delta_j - (x_j\beta - \alpha p_j)$$

Estimation

- ▶ GMM objective function:

$$f = \zeta(\theta)' Z \Phi^{-1} Z' \zeta(\theta),$$

where Φ^{-1} is a consistent estimate of $E[Z' \zeta \zeta' Z]$

- ▶ Search $\theta = (\alpha, \beta, \sigma)$ that minimizes the objective function
- ▶ Note: finding the nonlinear parameter σ requires a nonlinear search.

Problem set

Intermission: Newton's Method

- ▶ Idea: Approximate (univariate) nonlinear function $f(x)$ by its tangent at x_n , where $x_{n+1} = x_n + \epsilon$:

$$y = f'(x_n)(x_{n+1} - x_n) + f(x_n)$$

- ▶ Finding the root of this tangent is easy, set $y = 0$ and solve for x_{n+1} :

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- ▶ Iterating until $x_{n+1} \approx x_n$ yields the (local) root of $f(x)$
- ▶ Optimization: Apply Newton's method to $f'(x)$, then check $f''(x)$

Intermission: Newton's Method

```
# Newton's root-finding method
def newton_sqrt(a, tol=1e-12):

    if a < 0:
        raise ValueError("Number a should ...
                           be positive.")

    x_t = a / 2
    x_t1 = a / 2 - 1
    i = 0

    while abs(x_t1 - x_t) > tol:
        x_t = x_t1
        f_t = a - x_t**2
        f_t_derivative = -2 * x_t
        x_t1 = x_t - f_t / f_t_derivative
        i += 1
        print(f"Value at iteration {i} is ...
              {x_t1:.10f}.")

    return x_t1
```

Find $\sqrt{a}$, $\rightarrow$ find $x = \sqrt{a}$.

$$\rightarrow x^2 = a$$

$$\rightarrow f(x) = a - x^2 = 0$$

Newton's method is a contraction mapping if $f''(x)$ is continuous, $f'(x) \neq 0 \forall x \in \mathbb{R}$, and $q \in (0, 1)$ such that $|f(x)f''(x)| \leq q|f'(x)|^2 \forall x \in \mathbb{R}$